1.) AIM

To design and configure a network using packet Tracer and demonstrate the difference b/w RIP (Distance Vector) and OSPF (Link state) routing using the Same 3 router topology.

Procedure

Open Cisco Packet Tracer

Add 3 routers and connect them using appropriate cables

Add PCs to the routers

Assign IP addresses to all router interface and PCs.

First configure RIP using:

* router rip
* version 2
* network <network-address>

Test connectivity using ping.

check the routing table using show ip route.

Remove RIP configuration and configure OSPF using:

* router OSPF 1
* network <network-address> <wildcard-Mask> area 0

Again test using ping and check show ip route.

Compare RIP and OSPF

Sample Output

PC1 > ping 192.168.3.2

Reply from 192.168.3.2
Reply from 192.168.3.2
Reply from 192.168.3.2

Packets : Sent = 4 , Received = 4 , Lost = 0

Routing table :

RIP -> R

OSPF -> O

Result :

The network was successfully configured using RIP and OSPF. OSPF provides faster and more efficient route selection than RIP for ~~lar~~ larger networks.

2) AIM
To build a network with 4 Systems connected through a hub and separately through a Switch using packet Tracer and compare the collision and broadcast domains.

Procedure
open Packet Tracer
Add 4 PCs and one Hub
Connect all pcs to the hub
Assign IP addresses in the same network
Test connectivity using ping.
observe the communication between PCs.
Create another network using 4 PCs and one Switch
Connect all PCs to the switch
Assign IP addresses
This test connectivity and compare collision domain

Sample output

Device	Collision Domain	Broadcast Domain
hub + 4 PCs	1	1
Switch + 4 PCs	4	1

Result
The hub creates one collision domain, while the Switch creates separate collision domain for each port. Both have one broadcast domain When no VLANs are configured.

3. AIM

To capture and analyze FTP - control and FTP - Data connections separately using wireshark and understand their differences

Procedure

open Wireshark

Start capturing the network traffic

connect to an FTP server.

Perform FTP Login using username and password.

Transfer a file

stop the capture

Apply the filter :

ftp

observe the FTP - control connection

observe the FTP - Data connection

Compare their ports and purpose

Sample output

FTP - Control → TCP Port 21

FTP - Data → TCP Port 20

FTP - control : Used for commands such as login, directory listing and file requests.

FTP - Data : Used for transferring actual files/data.

Result

The FTP - control and FTP - Data connections were successfully captured and analyzed. Port 21 is mainly used for FTP control,

4. AIM

To design and configure a network for a retail chain having one head office and 5 Branch stores, where each branch communicates only with the head office using Packet Tracer.

Procedure

Open Cisco Packet Tracer

Create 1 Head office router

Create 5 branch routers.

connect each branch router to the head office router.

Add Pcs to each branch

Assign different IP networks to each branch.

Configure router interfaces.

Configure routing so that branches can communicate with the head office.

Test connectivity using ping

Verify that the network is working correctly

Sample Output

Head office Pc - Branch 1 Pc

Ping successful

Head office Pc - Branch 2 Pc

Ping successful

Head office Pc - Branch-3 Pc

Ping successful

Result :

Thus, a retail chain network with one head office and five branch stores was successfully designed and configured in cisco packet Tracer. The branch can communicate with the head office, while direct communication between branches is restricted.